



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Better connectivity and longer range for IoT devices

Synaptics Incorporated has introduced the Veros™ seamless intelligent connectivity brand, featuring system-on-chip (SoC) technologies for Wi-Fi

6E, Bluetooth, and 802.15.4. The SYN4383 continues Synaptics' 'triple combo' series, offering simultaneous dual-band

functionality and system integration. The SYN43756E updates the 2x2 Wi-Fi and Bluetooth combination SoCs with support for the 6E Wi-Fi band and LE Audio. In consumer electronics, manufacturers can use these chips in smart home devices and wearables to improve connectivity and battery efficiency. Automotive companies can use these SoCs in connected and autonomous vehicles for in-car Wi-Fi and vehicle-to-everything communications. Enterprises can integrate these systems into IoT solutions for automation and security in office and industrial environments.

Synaptics Incorporated
<https://www.synaptics.com>

First MPU with defence-grade security against future quantum threats

Microchip Technology has launched the PIC64HX family of microprocessors (MPUs). The product is designed for the needs of intelligent edge applications and differs from traditional MPUs by focusing on the specific requirements of these environments. The MPU, a multicore 64-bit RISC-V, is suitable for AI/

ML processing and includes integrated time-sensitive networking, Ethernet connectivity and post-quantum security. Aerospace manufacturers can deploy these MPUs for precise and reliable flight control and avionics, while

automotive manufacturers can utilise them for secure driver-assistance and infotainment systems. They are also ideal for industrial automation and real-time manufacturing processes.

Microchip Technology
<https://www.microchip.com>

Energy-efficient AI accelerator for edge applications

MemryX has launched the MX3 Edge AI Accelerator. It is a tool for applications like video management systems due to its high frame rate enabled by dataflow and at-memory computing architecture. The solution also offers accuracy, enabling the

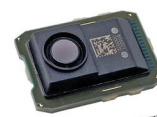


compilation and execution of thousands of AI models at a single click without retraining. It simplifies AI deployment through automated processing of pre and post tasks. The MX3 is scalable, functioning either individually or in combinations without additional hardware, adaptable from single-chip smart cameras to multi-chip edge servers. Each chip consumes 0.5 watts to 2.0 watts, supporting performance in fanless devices like industrial PCs, and uses less power than mainstream GPUs while delivering performance.

MemryX
<https://memryx.com>

Sensor for fast recognition of people and objects

Mitsubishi Electric Corporation has launched the MelDIR-brand 80 × 60 pixel thermal-diode infrared sensor



(MIR8060C1), which has a 100° × 73° field of view, more than doubling the viewing angle of its previous models.

This advancement will reduce the number of sensors needed for monitoring across large areas, improving solutions for elderly care facility monitoring, building air-conditioning systems, people counting, and body temperature measurement. The sensor features a design that minimises incident-light components, improving the clarity of thermal images and allowing for wide-area monitoring with fewer units.

Mitsubishi Electric Corporation
<https://www.mitsubishielectric.com/en>

Fast storage with more capacity in less space

Kioxia Corporation has launched the industry's first universal flash storage (UFS) Version 4.0 embedded flash memory devices using 4-bit-per-cell,



quadruple-level cell (QLC) technology. QLC UFS delivers a higher bit

density compared to traditional triple-level cell (TLC) UFS, making it suitable for mobile applications requiring larger storage capacities. It is particularly suitable for smartphones and tablets, and other cutting-edge applications like PCs, networking, augmented/virtual reality,